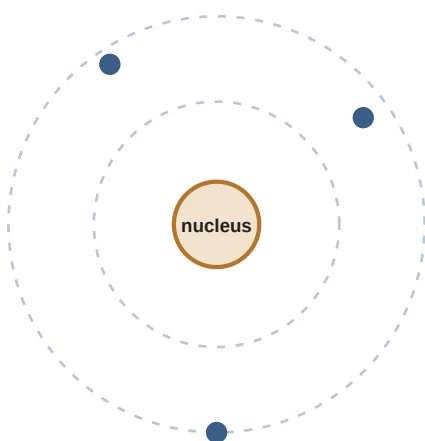


Atomic Structure

Quick reference sheet · Physics · keep handy for revision

1. What's inside an atom

Every atom has a tiny, dense **nucleus** (protons + neutrons) with **electrons** occupying shells (energy levels) around it. Atoms have no overall charge — the number of electrons always equals the number of protons.



Particle	Relative charge	Relative mass	Location
Proton	+1	1	Nucleus
Neutron	0	1	Nucleus
Electron	-1	very small (~1/2000 of a proton)	Shells

2. Atomic number, mass number & isotopes

Atomic number = number of protons (defines the element). **Mass number** = protons + neutrons.

protons = atomic number **electrons = atomic number** **neutrons = mass number – atomic number**

Isotopes: atoms of the same element (same atomic number) with a different number of neutrons (different mass number). Same chemistry, different mass.

3. Electron shells

Electrons fill the lowest available shell first. For GCSE (atomic number 1–20), shells fill in the pattern **2, 8, 8**.

Worked example: sodium, atomic number 11 → 11 electrons → fill 2, then 8, then what's left → **2, 8, 1**.

4. Worked examples

Given	Find	Working	Answer
Atomic no. 17, mass no. 35	Neutrons	$35 - 17$	18
Atomic no. 12	Electronic structure	$12 \rightarrow$ fill 2, 8, then remainder	2,8,2

Atomic no. 20, mass no. 40	Protons, neutrons, electrons	$p = 20, n = 40 - 20, e = 20$	20, 20, 20
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Tip: for "describe how the model changed" questions, always link the observation (what an experiment showed) to the conclusion (what it proved) — that link is usually where the marks are.